Heads Up Football Training Decreases Concussion Rates in High School Football Players

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Abstract

Objective: The objective of this study was to determine whether football players receiving heads up (HU) training would display a lower rate of sport-related concussion (SRC) compared to those without HU training. **Design:** Prospective cohort. **Setting:** Scholastic football fields. **Patients:** Two thousand five hundred fourteen scholastic football players during the 2015 to 2016 football season were monitored throughout the football season. **Interventions:** Before the preseason, 1 coach from 14 schools received HU training from US football. These schools were matched with 10 control schools performing standard football training [non–heads up (NHU)]. Random monitoring for proper coaching instruction was performed during the season. School athletic trainers monitored injuries and return to sport over the season for all schools. **Main Outcome Measures:** The outcome measures of concussions and associated time loss were determined before data collection began. Injury incidence and rate ratios were calculated to determine program effectiveness. **Results:** During the season, football players sustained 117 concussions (HU = 75; NHU = 42). The HU players displayed a significantly lower concussion rate (4.1 vs 6.0/100 players) compared with NHU teams [rate ratio = 1.5; 95% confidence interval (CI), 1.1-5.4]. The HU group returned to full participation 27% faster than athletes in the NHU group [time loss: 18.2 days (95% CI, 15.8-20.6) vs 24.8 days (95% CI, 19.9-29.7)], respectively. **Conclusions:** This is the first study to evaluate the impact of the HU program on the incidence of SRC in high school football players. Our data demonstrated that the HU program reduced SRC rates by 33% supporting the use of US HU football training as an effective method to decrease the rate of SRCs in scholastic football. **Clinical Relevance:** This study is the first to prospectively explore the effect of HU training on the incidence and recovery from SRC in high school football players.

Key Words: concussion, injury prevention, adolescents, football

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INTRODUCTION

More than 44 million high school students played scholastic sports and more than 1 million played scholastic football during the 2014 to 2015 academic year.[1] There are many positive health benefits associated with regular participation in sports including the ability to maintain optimal weight, development of coordinated movement patterns, improved academic performance, and reduction in the development of chronic diseases.[2–5] However, negative factors associated with increased participation include anxiety, performance and social pressure, and a potential for injury.[6] Recently, the impact and incidence of sport-related concussions (SRCs) in athletes, especially those participating in contact sports has become a public concern. Some authors have suggested that public fears over concussions in football may be associated with a decline in football participation.[7]

The concerns regarding SRCs have led to increased research investigating the incidence, potential causes, and severity of SRCs.[8,9] In addition, several sports and medical associations have suggested rule changes, prevention, and education programs to reduce the number, severity, and health impact of SRC.[10–16] There are few studies reporting on the effectiveness of these education and prevention efforts to reduce the incidence and severity of SRC.[7,16,17] To our knowledge, there are no efficacy evaluations of the US football heads up (HU) program on SRC incidence and recovery in high school football players. Our study compared SRC incidence and recovery in a cohort of high school football players who received US HU information and instruction (intervention-HU) with those high school football players who participated without the benefit of HU information and training [control–non-HU (NHU)]. All schools participated under the same rules and medical oversight for SRC diagnosis and management. Therefore, we hypothesized that the athletes on intervention teams (intervention-HU) would have a significantly lower incidence of SRC and less time loss to participation than the players participating on control (NHU) teams over one playing season.

METHODS

This study compares the rate and recovery of SRC between 14 high school football teams with 47 football teams (n = 1818; mean of 38 participating athletes/team) receiving US Football

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HU Certification Course sponsored by US football to 10 high schools with 22 participating football teams (NHU; n = 696; mean of 32 participating athletes/team) that did not receive HU training. The 14 high schools were within Greenville county, and the 10 comparison schools were from other adjacent schools that did not receive training. All schools received medical care within the Greenville Hospital System Athletic Training Network providing the ability to monitor injuries, days lost, and treatment received. Each school minimum medical coverage included a minimum of one full-time athletic trainer (AT) (all practices and games) and a fellowship-trained primary care sports medicine physician or orthopedic surgeon at all home football games. All athletes were deemed academically and medically eligible to participate before the start of the season. Institutional review board approval was obtained before data collection (Pro00054926).

Before the start of preseason football, at least one full-time assistant football coach (player safety coach) from each HU-trained school and all network ATs (regardless of group assignment) completed the US football safety standards training focusing on cardiac arrest awareness and response training, SRC recognition, heat and hydration preparedness, equipment fitting, and emphasizing technique training for blocking and tackling. All research team members from the study monitoring organization (CEROrtho) also completed the training program.

The HU-trained player safety coach educated all other football coaches within the organization (intervention-HU schools) on all aspects of player safety including tackling and blocking techniques. The player safety coach directed (with the supervising assistance of additional coaches) all contact tackling and blocking drills to ensure the use of proper HU techniques during football practices. The program was implemented at all levels of football at the trained schools from the first day of preseason practice through the last official game during the scholastic season. Control schools and athletes did not participate in any special training or additional focus on SRC prevention including blocking/tackling technique as verified by each school's AT. The scholastic AT in all schools (HU- and NHU-trained) educated coaches, parents, and athletes on the signs, symptoms, and potential risks of SRCs in each school's mandatory preseason sport meetings.

Football players in both groups (HU and NHU schools) were monitored for participation in practice, availability for games, and the occurrence of an SRC from the first day of practice throughout the last day of the 2015 to 2016 competitive football season. A football-related SRC injury was defined as any type of traumatic brain injury caused by a contact injury, or rapid movement of the head causing the head and brain to move rapidly back and forth that occurred during any scholastic sport team–sponsored activity.[18,19] In all schools, SRC injuries were documented by the AT including the date, mechanism/event of injury, treating medical provider, time loss, and demographic data within the AT System (Keffer Development Services, LLC, Grove City, PA). Each concussed athlete was referred to a board-certified sports medicine physician within the Greenville system for evaluation, diagnosis, and follow-up care. The physician documented the SRC using a combination of history and symptom reports, physical examination, and appropriate imaging, complemented by neurocognitive and balance assessments. All players were managed using a standard diagnosis and return to play process based on current clinical guidelines.[18,19] The return to play process was coordinated by the physician and the scholastic AT under the same guidelines in each school.[19]

A fidelity control rubric (Figures 1 and 2) was used to evaluate implementation[20] and adherence to the US football HU health and safety training principles within the HU schools by an independent Greenville research assistant. A fidelity audit schedule was used to ensure each participating school was assessed on a minimum of 3 occasions for compliance with HU techniques during all contact drills throughout each monitored practice session. During randomly selected team practices (n = 3; one time during the preseason, one time after the fourth game of the season, and once after the 8 game of the season) for each school, the ability of the safety coach to instruct proper performance of techniques during tackling and blocking drills were evaluated by a research team member throughout the teams practice. In addition, we assessed how well the athletes performed each drill with proper technique and the ability of the coach to provide appropriate instructional cues. The research team member assessed tackling and blocking drills to determine proper instruction and performance. The assessment was documented as a discrete variable (yes/no) to denote whether the drill techniques were appropriately instructed and performed during the practice.[21] The number of steps within a drill that were correctly instructed and separately the number of steps within a drill that were correctly performed was tallied for each team (Table 1).

The team was deemed to have been compliant with HU training if the drills were both instructed and performed correctly at a minimum of 80% frequency within the practice. If the 80% threshold was not attained and maintained an additional fidelity, control session was scheduled within 14 days of the previous session. At the end of each evaluated practice, the coach and researcher discussed drill instruction, performance, and player feedback to determine effectiveness and potential modifications to the implementation.

Data Analysis

Mean values and SDs were calculated for all demographic and descriptive variables to describe the population of interest. Cumulative incidence and incidence rate ratios of SRCs were calculated with 95% confidence intervals (CIs) to determine program effectiveness. In addition, game and practice cumulative incidence rates and rate ratios were calculated with 95% CIs to determine within- and between-group differences. Specific practice incidence rates with rate ratios were calculated. Rate ratios (RRs) with 95% CIs not crossing 1.00 were considered statistically significant. Time loss was compared between groups using a 1-way analysis of variance (ANOVA) as a measure of SRC severity.

Ethical Considerations

Athletes in 10 of the 24 high schools were included as controls to determine the potential effect of HU training. Although it is necessary to provide best available interventions to those that would benefit, there is limited research to support the efficacy of this training. It was, therefore, necessary to follow a group of football teams that had not pursued participation in HU training to determine the potential benefit of the program.

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School		
Level of play		
Date		
Coach		
Instruction	Yes	No
Correctly explains goals of drill		
Correctly demonstrates technique		
Provides opportunity for questions		
Provides feedback and cues		
Tackling	Yes	No
Breakdown position		
•Cues proper athletic position		
•Corrects techniques		
	Yes	No
Foot movement		
•Cues proper controlled movement and position		
•Corrects techniques		
	Yes	No
Hit position		
•Cues proper controlled movement, heads up position, strike with shoulder		
•Corrects techniques		
	Yes	No
Shoot		
•Power through hip movement, heads up position		
•Corrects techniques		
	Yes	No
RIP		
•Head up, to the side and away from contact, symmetrical arm motion to wrap up tackle		
•Corrects techniques		
Breakdown position—foundation starting point for all activity. 45 degree trunk lean. Knees flexed. Weight on balls of feet. Buzzing the Feet— technique regains breakdown position prior to contact. Making sure you come in under control. Hit position— correct body posture at impact for safer tackling. Head and eyes up using front of shoulder as point of contact. The shoot— the opening of the hips to generate power and create an ascending tackle RIP-head to the side and away from contact, throw double uppercuts, and grab cloth on the back of jersey to secure tackle.		

Figure 1. Fidelity control audit form—tackling.

Two researchers were blinded to athlete, school, team, and group identity during data management to reduce the risk of bias.

RESULTS

A total of 117 SRCs were observed in the 2514 high school football players competing during the 2015 to 2016 competitive football season in the Upstate of South Carolina. All HU teams were deemed to have been compliant with the training with 80% or more tackling and blocking drills instructed and performed correctly at 3 randomly audited practices. Football players in the HU training cohort were similar in age (15.1 ± 1.4 vs 14.8 ± 1.6 y.o.; $P = 0.23$) and height (174.3 ± 8.1 vs 171.5 ± 10.4 cm; $P = 0.1$) but significantly heavier (80.4 ± 20.2 vs 69.8 ± 20.7 kg; $P = 0.008$) than players in the NHU cohort. Players on the HU training teams sustained 75 SRCs compared with 42 SRCs on NHU teams.

The HU rate of 4.1 SRCs/100 players (95% CI, 3.3-5.1) was significantly lower than the 6.0 SRC rate on NHU teams (RR = 1.5; 95% CI, 1.1-2.1) (Table 2). The number of cumulative SRCs in the entire cohort was higher in games (n = 61) versus practice (n = 56); the rate ratio (1.1; 95% CI, 0.73-1.5), however, was not significant. The SRC rate in games trended toward being higher in the NHU compared with the HU cohort (Table 2). The

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School		
Level of Play		
Date		
Coach		
Instruction	**Yes**	**No**
Correctly explains goals of drill		
Correctly demonstrates technique		
Provides opportunity for questions		
Provides feedback and cues		
Blocking	**Yes**	**No**
Stance position		
•Cues proper athletic position (chin up, eyes open, feet shoulder width, knees bent, and back flat) Weight on balls of feet		
•Corrects techniques		
	Yes	**No**
Get off		
•Cues proper controlled movement and position (eyes up, hands to chest, 2 feet on ground, back flat)		
•Corrects techniques		
	Yes	**No**
Strike position		
•Cues proper controlled movement, heads up position, strike with both hands upward		
•Corrects techniques		
	Yes	**No**
Stick		
•Power through hip movement, heads up position		
•Corrects techniques		
	Yes	**No**
Finish		
•Head up, wide base of support , movement in control, connected to defender		
•Corrects techniques		
Breakdown position—foundation starting point for all movement. 45 degree trunk lean. Knees flexed. Weight on balls of feet. Buzzing the feet— technique for coming to balance and regaining breakdown position prior to contact. Making sure you come in under control. Hit position— correct body posture at moment of impact for safer tackling. Head and eyes are up using front of shoulder as point of contact. The shoot— the opening of the hips to generate power and create an ascending tackle RIP- with head to the side and away from contact, throw double uppercuts, and grab cloth on the back of jersey to secure tackle.		

Figure 2. Fidelity control audit form—blocking.

specific incidence of practice SRCs was significantly greater in the NHU versus the HU group (RR = 1.9; 95% CI, 1.1-3.2) (Table 3). The HU football group returned to full participation 27% faster than athletes in the NHU football group [days loss: 18.2 ± 10.6 days (95% CI, 15.8-20.6) versus 24.8 ± 15.3 days (95% CI, 19.9-29.7, respectively)].

DISCUSSION

Our study shows that the risk of an SRC was 1.5×'s or 32% lower for high school football players from HU-trained schools compared with the cohort of untrained schools (4.1 vs 6.0/100 players). In addition, the specific injury rate during practices was greater in the NHU versus the HU group

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TABLE 1. Example of Compliance Audit Compliance Implementation Sheet

			Session Success: ☑ Yes ☐ No			
Team:			Date: 8/18/2015			
Coach:			Feedback: 8/18/2015			
Tackling	**Breakdown position**	**Foot movement**	**Hit position**	**Shoot**	**RIP**	**Total**
Instruction	26/26	26/26	26/26	26/26	26/26	130/130 (100%)
Performance	24/26	23/26	29/26	21/26	25/26	112/130 (86%)
Each item from the HU checklist (Figure 1) was scored during blocking and tackling drills and then summarized to generate an adherence total percentage of correct performance or instruction/participating athletes						
Blocking	**Stance position**	**Get off**	**Strike position**	**Stick**	**Finish**	**Total**
Instruction	26/26	0/26	26/26	26/26	26/26	105/130 (81%)
Performance	26/26	21/26	22/26	24/26	23/26	116/130 (89%)
Each item from HU checklist (Figures 1 and 2) was scored during tackling and blocking drills and then summarized to generate correct performance adherence percentage of performance and participation.						

indicating that the benefits of this training might be most effective in practice. These results suggest that US football's HU program is effective to lower the overall rate of SRCs in high school football players. The US football's HU program is promoted as a method to provide a better, safer experience for young football players and their families. This study provided the opportunity to compare the SRC rates between a cohort of football players coached by HU football–trained coaches to a cohort of players who continued their standard coaching.

To our knowledge, the only comparable study was recently published by Kerr et al[7] examining SRCs in youth football athletes from 5 to 15 years of age (mean = 10.8). The cumulative SRC rate in their cohort was 2.1 SRCs/100 players, which is lower than in our cohort (4.7 SRCs/100 players). The difference in SRC rates between the 2 cohorts was likely related to differences in age groups as well as contact rule differences between high school versus pop warner leagues. However, similar to our study, the rates of SRCs were greater in the NHU athletes as compared to pop warner HU-trained athletes.[7] The overall SRC rate in our cohort of all football players was 4.7 SRCs/100 football players, which was slightly lower overall than recently reported rates.[22] However, the frequency of SRCs in the untrained cohort was similar to the incidence reported in other cohorts of high school football players who also had staff ATs to verify the occurrence of injury.[8,22]

To compare our rates to data reported in the literature, we calculated cumulative incidence from a cohort of high school football players presented by Dompier et al.[8] The cumulative practice SRC incidence rate in the high school football players in the aforementioned studies would be 2.8 per 100 players.[8,22] This rate would be comparable with the incidence in our NHU athletes. We were able to get detailed athlete exposure for practices but not games in this large cohort, based on the geographic spread of the schools, the multiple levels of game action (varsity, junior varsity, and lower levels), and limited resources. However, there were no significant differences among teams in terms of SRC rates, suggesting that using team exposures was adequate. The interpretation of this information requires caution and is limited by the lack of ability to account for the specific player participation in each football game, but all teams were under the same state guidelines for total number of practices and games.

Heads up–trained football players returned to full participation almost 1 week faster [time loss: HU = 18.2 ± 10.6 days (95% CI, 15.8-20.6) versus NHU = 24.8 ± 15.3 days (95% CI, 19.9-29.7)] compared with the NHU players. All players were treated and managed using a standard diagnosis and return to play protocol based on current clinical guidelines.[18,19] Thus, the differences in time to return to sport is in part a reflection of the rate of symptom resolution and the ability of the athlete to complete a graded program to resume academic and athletic activities without the recurrent symptoms. The HU football group returned to full participation 27% faster than athletes in the NHU football group under the same medical oversight and guidelines.

TABLE 2. Cumulative Injury Rates for Concussion in NHU-Trained Versus HU-Trained Football Players

	Total			NHU			HU				
Injury	**N***	**Athletes†**	**Rate‡**	**N***	**A†**	**Rate‡**	**N***	**Athletes†**	**Rate‡**	**Rate Ratio**	**(95% CI)**
SRC rate	117	2514	4.7	42	696	6.0	75	1818	4.1	1.5	(1.1-5.4)
Practice SRC rate	56	2514	2.2	19	696	2.7	37	1818	2.0	1.34	(0.77-2.3)
Game SRC rate	59	2514	2.4	23	696	3.3	36	1818	2.0	1.67	(1.0-2.8)
Practice game rate ratio			1.1			1.2			0.97		
(95% CI)			(0.73-1.5)			(0.67-2.2)			(0.61-1.5)		

*N = number.
† Athletes = participants with exposure to risk.
‡ Rate = injury rate per 100 athletes.

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TABLE 3. Specific Practice Concussion Incidence in NHU Compared With HU

	Total			NHU			HU				
Injury	N*	AEs†	Rate‡	N*	AEs†	Rate‡	N*	AE†	Rate‡	Rate Ratio	(95% CI)
Practice SRC rate	56	1 799 490	3.1	19	386 447	4.9	37	1 413 043	2.6	1.9	(1.1-3.2)

*N = number of injuries.
† AEs = each time a player participated in a practice.
‡ Rate = injury rate per 100 000 athlete exposures.

The observed lower rate of SRCs and shorter duration of time loss in players using HU techniques when compared to those without modification of contact drills may represent an ability to lower the risk of SRC by monitoring and improving player skills. The specific rate of practice SRCs was higher in the NHU players as compared to HU players (RR = 1.9; 95% CI, 1.1-3.2), but the cumulative rate of SRC's (RR = 1.34; 95% CI, 0.77-2.3) difference did not reach statistical significance. The fidelity monitoring confirmed that athletes were following recommended instruction, blocking and tackling techniques aimed to limit head contacts by instructing proper blocking and tackling techniques during practice. Our results suggest based on the lower rate and apparent severity that this approach was effective.

Strengths of the Study

The study's design established that all athletes were without SRC symptoms before participation in the scholastic football season. Fidelity testing was performed randomly at all participating intervention schools to ensure consistent training. Recall bias and confirmation of injuries were minimized as staff ATs collected the data in a systematic, prospective manner. The study was conducted over a small geographical area, which allowed us to control factors related to rules of play, schedules of teams (climate and competition level), and documentation and management of athlete SRCs. Practice exposures were available and recorded by each school's athletic training staff allowing for a comparison of specific and normalized practice injuries by athletic exposure (AE). The primary providers for these athletes (both ATs and physicians) are part of the same large health network ensuring that the same clearance and 6-stage return to activity program were used to return athletes to play regardless of training group.

Limitations

Our results should be viewed within existing limitations. The assignment to HU training group was not randomized. The groups were formulated based on availability of training, and therefore, the groups represent an unequal number of athletes. The injury rates were normalized to the number of participants per group, but the lack of randomization to groups may introduce some heterogeneity and bias between groups. Examples of these potential biases include the following:

1. Athletes in the HU group tended to be heavier but not taller or older.
2. The schools in the NHU (control group) were on average smaller based on classification. Although all schools regardless of group assignment scheduled the same number of games during the regular season and similar numbers of players per team, exact game exposures were not available to create specific game incidence rates leaving less precise cumulative rates for comparison. The monitoring of game participation was prohibitive based on the availability of resources because many of our athletes play at multiple levels and play a disproportionate number of plays during game competition. However, the cumulative rates create a more conservative approach compared with the more precise exposure and specific incidence rates. Thus, our game estimates were likely more conservative compared to an analysis with detailed exposure data providing the most conservative estimate of effect of the training group. Finally, we were not able to get detailed position data because this is prohibitive in high school football with players involved in multiple phases of the game. Although it was likely that position influenced the rate of SRCs among groups, there is no reason to believe that the groups would be different in their distribution of SRCs across positions between groups. However, this limitation suggests important next steps in understanding scalable interventions to decrease the SRC injury risk. It is possible that there were differences between groups potentially escalating the appearance of practice risk over the season. In summary, the major study design limitations include lack of randomization, unequal sample size, and an inability to collect game exposures by play and position. These limitations may impact the overall findings, and the study will require replication to confirm our results. However, these findings are an important first step in understanding the potential role that the HU program may provide in high school football.

CONCLUSIONS

There is an inherent risk of suffering SRC while participating in football. Improved awareness through programs such as US HU football may refine blocking and tackling skills helping to protect athletes from injury. Our results should be interpreted with caution because of our small and unequal groups. The inability to account for game exposures and position of injury limit the generalizability of our results. Our initial results suggest that HU training may assist in lowering the risk of SRC (33% in practice, 40% in games, and 27% in time loss) in scholastic football players. These results require confirmation in a larger and more well-controlled study performed over a longer period.

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